

Montaser Mohammadalamen, PhD

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WORK EXPERIENCE

• Alberta Machine Intelligence Institute (Amii)

July 2025 - Present

Applied Research Scientist, AI Trust & Safety

Edmonton, Canada

- **Research Strategy:** Establishing Amii's safety research dual-track strategy using RL to risk assignment generative models, and mitigating structural RL safety (uncertainty, non-stationarity, partial reward observability).
- **Technical Collaboration:** Partnering with Canadian AI Safety Institute (CAISI), Canadian Institute for Advanced Research (CIFAR), and national institutes, Mila, the Vector Institute, and the National Research Council (NRC).
- **Academic Outreach:** Directed a 10-session technical AI safety track for Amii's annual Upper Bound conference, serving an audience of over 11,000 attendees.
- **Team Building:** Hired a Research Scientist and an ML Engineer for the team while overhauling the technical interviews framework now adopted by Amii.

• SonyAI

June 2020 - Oct 2020, Jan 2021 - Oct 2022

AI Engineer

Tokyo, Japan

- **Robotics Research:** Co-developed a table-tennis robot on the ACE Team capable of defeating professional players, resulting in a [Nature](#) publication.
- **RL Policy:** Trained RL policies for targeted, high-precision ball returns and led the technical sub-team for robot serving.
- **Sim-to-Real & System Integration:** Built multi-agent simulation environments using self-play and goal-conditioned RL, successfully transferring trained RL policies onto real-time robot control loops.

• University of Alberta

Feb 2020 - Dec 2020

Research Associate

Edmonton, Canada

- Formulated a novel robust optimization framework that enables autonomous control agents to learn cautious behaviors when encountering novel, out-of-distribution states, supervised by Prof. Michael Bowling.

• SonyAI

July 2019 - Jan 2020

Machine Learning Intern

Tokyo, Japan

- Design an RL environment and trained policies using imitation learning to acquire complex cooking skills from chef demonstrations, successfully deploying the models on physical robots.

EDUCATION

• University of Alberta — PhD in Computing Science

Sep 2022 – Dec 2025

- **Thesis:** [Reinforcement Learning with Partially Observable Rewards](#).
- **Advisor:** [Prof. Michael Bowling](#)

• African Institute for Mathematical Sciences — M.Sc. in Machine Intelligence

Sep 2018 – Sep 2019

- **Performance:** Overall score 75%
- **Advisor:** [Prof. Benjamin Rosman](#)

• University of Khartoum — B.Sc. (Honors) in Electronics & Computer Engineering

Aug 2013 – Sep 2018

- **Performance:** First Class Honors (Ranked 3rd in the Electronics department, CGPA 7.5/10)

SELECTED PUBLICATIONS & PATENTS

Peer-Reviewed Journal Articles

- [J.1] **Montaser Mohammedalamen**, Dustin Morrill, Alexander Sieusahai, Yash Satsangi, and Michael Bowling, “[Learning to Be Cautious](#)”, *Transactions on Machine Learning Research (TMLR)*, 2025.

Refereed Conference Publications

- [C.1] **Montaser Mohammedalamen** and Michael Bowling, “[Generalization in Monitored Markov Decision Processes \(Mon-MDPs\)](#),” *Reinforcement Learning Conference (RLC)*, Montreal, Canada, 2026.
- [C.2] Simone Parisi, **Montaser Mohammedalamen**, Alireza Kazemipour, Matthew E. Taylor, and Michael Bowling, “[Monitored Markov Decision Processes](#)”, *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, Auckland, New Zealand, 2024.

Workshop Papers & Preprints

- [W.1] **Montaser Mohammedalamen**, K. Roice, R. McLean, A. Lefaivre Škopac, “[A Systematic Investigation of The RL-Jailbreaker in LLMs](#),” *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Machine Learning (ICML)*, 2026.
- [W.2] R. McLean, T. E. Lee, **Montaser Mohammedalamen**, K. Roice, G. Berseth, P. M. Pilarski, M. C. Machado, A. Lefaivre Škopac, B. Rosman, “[AI Agent Safety is a Reinforcement Learning Problem](#)”, *Agents in the Wild: Safety, Security, and Beyond Workshop, International Conference on Machine Learning (ICML)*, 2026.
- [S.1] **Montaser Mohammedalamen**, Michael Bowling, and Matthew E. Taylor, “Learning To Be Cautious for Industrial Chemical Control,” *Under review*.
- [W.3] D. Khamies, **Montaser Mohammedalamen**, and B. Rosman, “[Transfer Learning for Prosthetics Using Imitation Learning](#),” *Black in AI workshop, NeurIPS*, arXiv preprint 1901.04772, 2018.

Theses

- [T.1] **Montaser Mohammedalamen**, “[Reinforcement Learning with Partially Observable Rewards](#)”, PhD Thesis, University of Alberta, 2026.
- [T.2] **Montaser Mohammedalamen**, “Learning Actions Representation In Reinforcement Learning for Safe Exploration,” Master’s Thesis, 2019.

PROFESSIONAL SERVICE

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| • Transactions on Machine Learning Research (TMLR)
<i>Journal Reviewer</i> | 2024 – Present |
| • International Conference on Machine Learning (ICML)
<i>Gold Reviewer</i> | 2026 |
| • Neural Information Processing Systems (NeurIPS)
<i>Conference Reviewer</i> | 2025 & 2026 |
| • Reinforcement Learning Conference (RLC)
<i>Technical Reviewer</i> | 2025 & 2026 |
| • Adaptive and Learning Agents (ALA) Workshop at AAMAS
<i>Workshop Co-organizer and Area Chair</i> | 2025 & 2026 |
| • Amii Fellows Trust & Safety Research Proposals
<i>Committee Member</i> | 2026 |
| • Cisco-University of Alberta Consortium for Agentic Research (CUACAR)
<i>Research Proposal Reviewer</i> | 2026 |

TEACHING EXPERIENCE

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|---|---|
| • University of Alberta
<i>Head Teaching Assistant — Experimental Mobile Robotics (COMP 412/503)</i> | Jan 2023 – May 2025
Edmonton, Canada |
| ◦ Coordinated the TA team, managed grading frameworks, and oversaw lab infrastructure for a mixed cohort of senior undergraduate and graduate students. | |
| ◦ Designed and developed 8 comprehensive lab modules for 35 students using the Duckietown to demonstrate physical robotic control. | |

- Instructed students on core robotics principles, including basic kinematics, ROS commands, P/PD/PID controllers, computer vision integration, and ML-based traffic sign detection.
- Mentored students through a capstone final project implementing an end-to-end autonomous driving pipeline featuring lane tracking, obstacle avoidance, and traffic rule navigation.
- Delivered specialized guest lectures on RL in robotics alongside career development strategies for industry internships.

SELECTED PROJECTS

• RL for LLMs Risk Assessment

Ongoing

Generative models Safety

- Tech Lead, developing a three-stage optimization framework that reframes generative model jailbreaking as a sequential decision problem. The system structurally decomposes environmental and algorithmic components to isolate the core drivers of adversarial success, scales exploitation pipelines across multi-modal action spaces (text, vision, and audio), and embeds these insights into a unified two-player zero-sum game.

• Learning to Be Cautious for Industrial Chemical Control

Ongoing

Critical Applications Safety

- Generalized the “learning to be cautious” framework to continuous action spaces, extending its mathematical applicability to high-dimensional, safety-critical settings, enabling robust control under environmental uncertainty. Validated the approach on complex chemical process control benchmarks, demonstrating practical scalability and bridging the gap between theoretical safe RL and real-world industrial deployments.

• Plasticity Loss in Monitored MDPs

Ongoing

Continual RL

- Investigating the structural mechanisms of plasticity loss within Monitored Markov Decision Processes (Mon-MDPs) to preserve agent adaptability during continual learning. Analyzing how the dual-learning architecture, simultaneously training a predictive reward model alongside deep Q-networks, compounds drives long-term performance degradation. Formulating a systematic framework to evaluate the intrinsic vulnerabilities of Mon-MDPs to this phenomenon while adapting advanced regularization and reset mitigation strategies to maintain an optimal stability-plasticity balance.

• Monitored Markov Decision Processes

Theoretical & Practical Evaluation frameworks



- Addressed settings where reward structures are unobservable to the agent, analyzing consequences across varied testing scenarios.

• Learning to be Cautious

Robust optimization algorithms



- Constructed parameters enabling systems to safely deploy context-aware behavior patterns dynamically within unfamiliar boundaries.

SKILLS

- **Programming Languages:** Python, JAX, PyTorch, TensorFlow, C++
- **Machine Learning & RL:** Reinforcement Learning (RL), Safety & Trust, Multi-agent RL, Optimization, Imitation Learning, Generative models, LLMs safety, Jailbreaking LLMs, Game Theory, Continual RL.
- **Deep Learning Architectures:** CNN, RNN, LSTM
- **Robotics & Deployment:** Robotics, ROS, Physics Simulation, Linux, Git, LaTeX, AWS, GCD, Slurm

PROJECT MANAGEMENT

• Professional Scrum Product Owner (PSPO I)

Certified

Scrum.org



• Project Management Professional (PMP)

Exam Pending

Project Management Institute (PMI)



HONORS AND AWARDS

- **Andrew Stewart Memorial Graduate Prize** May 2025
University of Alberta (5,000 CAD)
- **Graduate Student Engagement Scholarship** Jan 2024
University of Alberta (10,000 CAD)
- **Graduate Student Engagement Scholarship** Jan 2023
University of Alberta (10,000 CAD)
- **AMMI Master Scholarship** Sep 2019
African Master in Machine Intelligence (15,000 USD)
- **Best Graduation Project ("RL for Prosthetics")** Oct 2018
University of Khartoum, Sudan
- **Undergraduate Research Prize** Dec 2017
University of Khartoum (2,000 USD)
- **Audience Prize** Nov 2017
Falling Walls Lab Finals, Berlin, Germany
- **ICT Professional Foundation Scholarship** Nov 2016
Ericsson Middle East University Program
- **IEEE Extreme 10.0 Programming Competition** Oct 2016
IEEE Sudan Section Ranked 5th in Sudan, 988th of 2500 teams globally
- **Best Project Award** 2015 - 2016
Electrical & Electronics Engineering Students Exhibition Awarded consecutively for two years

VOLUNTEERING

- **Sudanese Machine Learning Community (SMLC)** Dec 2019 - Present
Co-founder & Core Representative 
 - Co-founded an initiative building AI capacity locally, connecting resources to regional engineering candidates.
- **SMLC Mentorship Program** Dec 2019 - Present
Academic Research Mentor
 - Mentoring undergraduate students at the University of Khartoum on custom baseline RL graduation items.
- **EEESE Hackathon** Apr 2016 - Aug 2016
Founder & Event Coordinator
 - Established the first hackathon event setup in Sudan, coordinating 28 student researchers from 7 distinct universities.
- **Academic Committee for EEESE** Aug 2015 - July 2016
Head of Committee University of Khartoum